

RESEARCH AND DOCUMENTATION

1.What is cybersecurity, and why is it important?

The practice of protecting digital information, network system and technologies from unauthorized access, use disruption, modification or destruction .

Importance:-

- This include safeguarding against various cyber threats, such as **hacking, malware, phishing** etc.
- Protecting the confidentiality , integrity, and availability of digital information.

2. 5 real world cyberattacks and how they happened.

1. **Yahoo data breach (2013-14)**:-,vulnerability in yahoo security question feature.
2. **Wannacry Ransomware Attack (2017)**:- phishing email with malware attachment.
3. **Target Data Breach (2013)**:-, malware is installed on point-of-sale system via phishing email.
4. **Equifax Data Breach (2017)**:- unpatched software allowed hacker to access sensitive data.
5. **Sony picture hack (2014)**:- phishing email with malware attachment.

3.Difference between HTTP and HTTPS

- **HTTP(Hypertext Transfeer Protocol)**: unsecured, transmits , data in plaintext , susceptible to interception.
- **HTTPS (HTTP Secure)**: secured with SSL/TLS encryption , ensure data confidentiality and authenticity .

4.Explanation of AES and RSA encryption with example

- **AES(Advance Encryption Standard)**: symmetric encryption (same key for encryption and decryption)
 - **Ex:** encryption “HELLO” with key 123 produces unreadable text.
- **RSA (Rivest-Shamir-Adleman)**: asymmetric encryption (user public and private keys.)
 - **Ex:-** public keys encrypted “HELLO” only the private kry decrypts it back to the original.

